

# Module 9: Outlier Detection

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- ❑ Basic concepts
- ❑ Statistical approaches
- ❑ Proximity-based approaches
- ❑ Reconstruction-based approaches
- ❑ Clustering and classification based approaches
- ❑ Mining contextual and collective outliers
- ❑ Outlier detection in high-dimensional data
  - ❑ Extending Conventional Outlier Detection
  - ❑ Finding Outliers in Subspaces
  - ❑ Outlier Detection Ensemble
  - ❑ Taming High-Dimensionality by Deep Learning
  - ❑ Modeling High-Dimensional Outliers

# Challenges: Outlier Detection in High-Dimensional Data

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- ❑ Interpretation of outliers
- ❑ Data sparsity
  - ❑ The distance between objects becomes heavily dominated by noise as the dimensionality increases
- ❑ Adaptive to data subspaces
- ❑ Scalability with respect to dimensionality

# Extending Conventional Outlier Detection

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- ❑ Use the conventional proximity-based models of outliers
- ❑ To overcome the deterioration of proximity measures in high-dimensional spaces, use alternative measures or construct subspaces and detects outliers there
- ❑ HiOut finds distance-based outliers, but uses the ranks of distance instead of the absolute distance in outlier detection
  - ❑ Find the k-nearest neighbors of  $o$ , denoted by  $nn_1(o), \dots, nn_k(o)$ , where  $k$  is an application-dependent parameter
  - ❑ The weight of object  $o$  is  $w(o) = \sum_{i=1}^k dist(o, nn_i(o))$
  - ❑ All objects are ranked in weight-descending order
  - ❑ The top- $l$  objects in weight are output as outliers, where  $l$  is another user-specified parameter